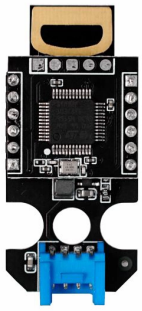
Unit UWB

SKU:U100



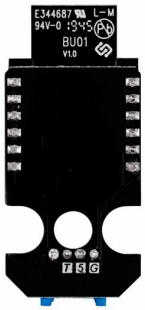
















Description

Unit UWB is a wireless communication unit equipped with indoor positioning technology. The design adopts the Ai-Thinker BU01 module solution (an ultra-wideband (UWB) transceiver module based on Decawave's DW1000). It features a built-in STM32 and integrated ranging algorithm, achieving a positioning accuracy of up to 10 cm, and supports AT command control. When applied to indoor wireless ranging, it operates in base station and tag modes (the base station calculates and outputs position information to the tag).

Features

- Positioning accuracy: 10 cm
- Built-in STM32 with integrated ranging algorithm
- AT command control
- Serial communication (baud rate: 115200)
- Easy integration, no RF design required
- Complies with IEEE 802.15.4-2011 UWB standard
- Supports two-way ranging and TDOA
- Development platforms: Arduino, UIFlow (Blockly, Python)
- 2 x LEGO-compatible holes

Includes

- 1 x Unit UWB
- 1 x HY2.0-4P Grove cable (20 cm)

Applications

- Indoor positioning/wireless ranging

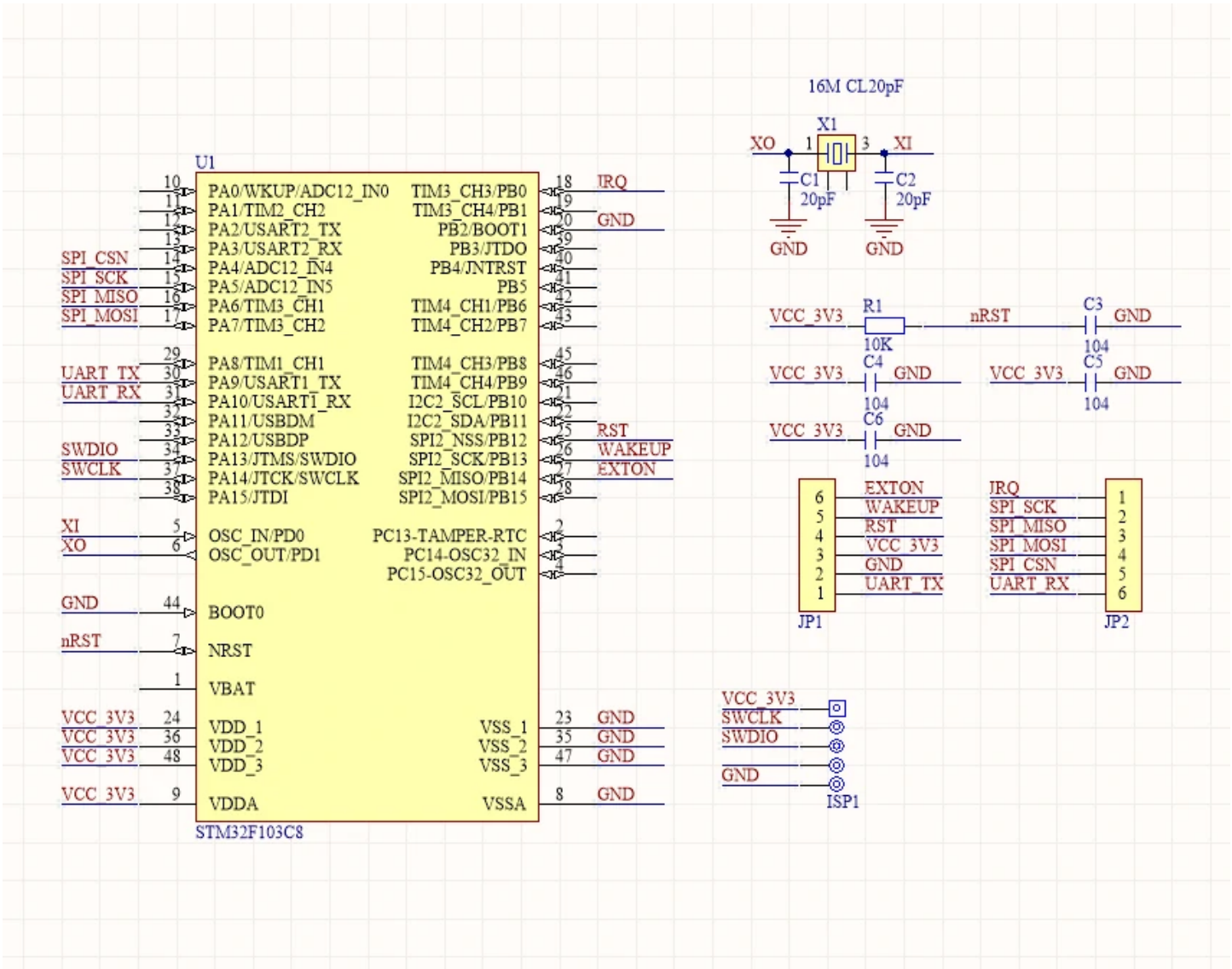
Specifications

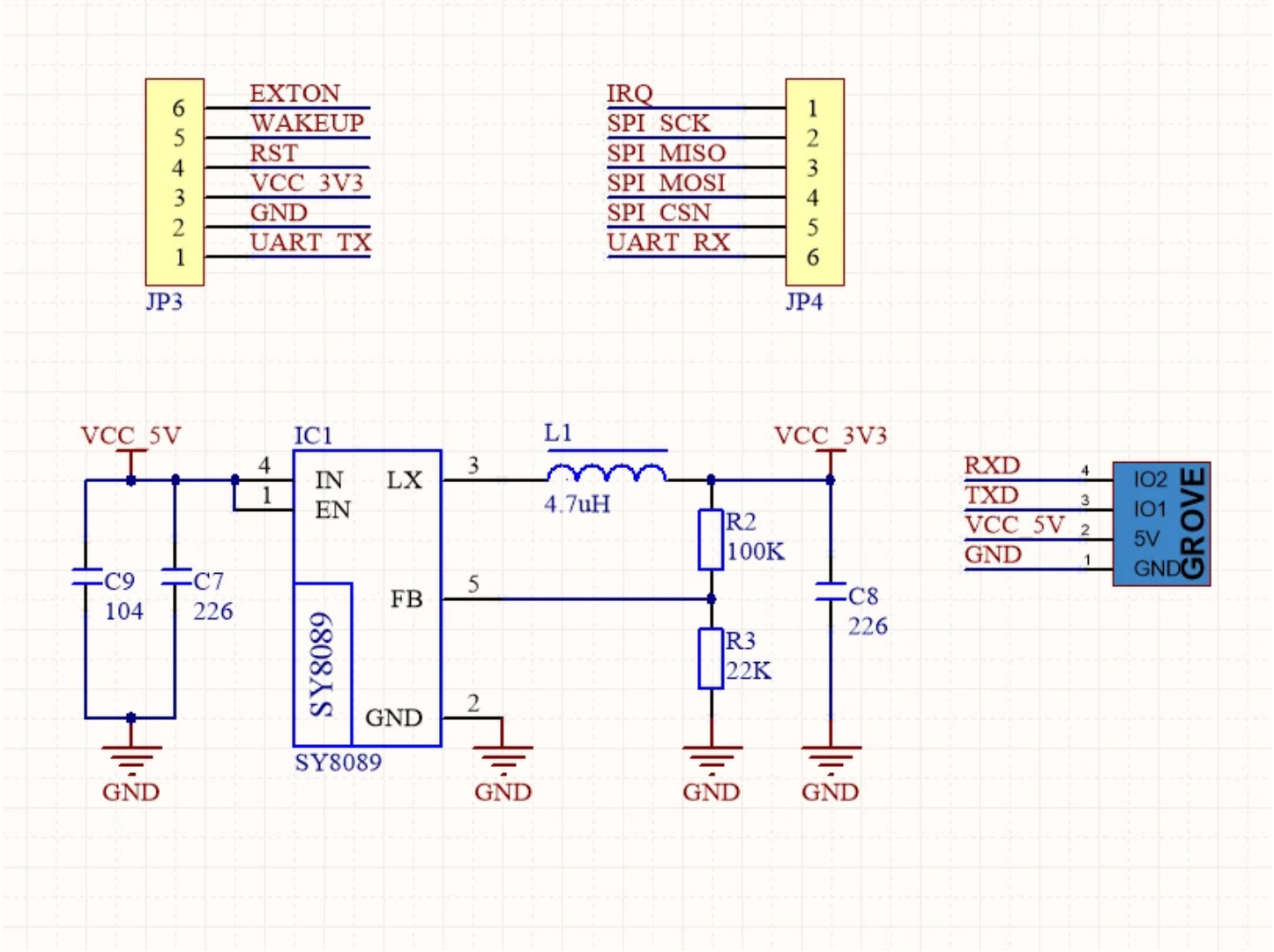
Specification	Parameter
MCU	STM32F103C8T6
Data transfer rate	110 kbit/s, 850 kbit/s, and 6.8 Mbit/s
Frequency band	6 bands: from 3.5 GHz to 6.5 GHz
Transmit power	-14 dBm/-10 dBm
Transmit power density	<-41.3 dBm / MHz
Supported packet size	1023 bytes
Modulation method	BPM (Binary Phase Modulation) and BPSK (Binary Phase Shift Keying)
FDMA	6 channels
Product Size	48.0 x 24.0 x 8.0mm
Product Weight	8.2g
Package Size	138.0 x 93.0 x 9.0mm
Gross Weight	13.5g

Learn

Note: The firmware currently installed on this unit only supports the transmission of ranging information and does not support custom information transmission. When in use, it supports the configuration of up to 4 base station devices (using different IDs), and only a single tag device is allowed to operate at any given time.

Schematics


U1
PA0/WKUP/ADC12_IN0
PA1/TIM2_CH2
PA2/USART2_TX
PA3/USART2_RX
PA4/ADC12_IN4
PA5/ADC12_IN5
PA6/TIM3_CH1
PA7/TIM3_CH2
PA8/TIM1_CH1
PA9/USART1_TX
PA10/USART1_RX
PA11/USBDM
PA12/USBDP
PA13/JTMS/SWDIO
PA14/JTCK/SWCLK
PA15/JTDI
OSC_IN/PD0
OSC_OUT/PD1
BOOT0
NRST
VBAT
VDD_1
VDD_2
VDD_3
VDDA
TIM3_CH3/PB0
TIM3_CH4/PB1
PB2/BOOT1
PB3/JTDO
PB4/JNTRST
PB5
TIM4_CH1/PB6
TIM4_CH2/PB7
TIM4_CH3/PB8
TIM4_CH4/PB9
I2C2_SCL/PB10
I2C2_SDA/PB11
SPI2_NSS/PB12
SPI2_SCK/PB13
SPI2_MISO/PB14
SPI2_MOSI/PB15
PC13-TAMPER-RTC
PC14-OSC32_IN
PC15-OSC32_OUT
VSS_1
VSS_2
VSS_3
VSSA
STM32F103C8
SPI_CSN
SPI_SCK
SPI_MISO
SPI_MOSI
UART_TX
UART_RX
SWDIO
SWCLK
XI
XO
GND
nRST
VCC_3V3
IRQ
RST
WAKEUP
EXTON
16M CL20pF
X1
C1
20pF
C2
20pF
R1
10K
C3
104
C4
104
C5
104
C6
104
JP1
JP2
ISP1



PinMap

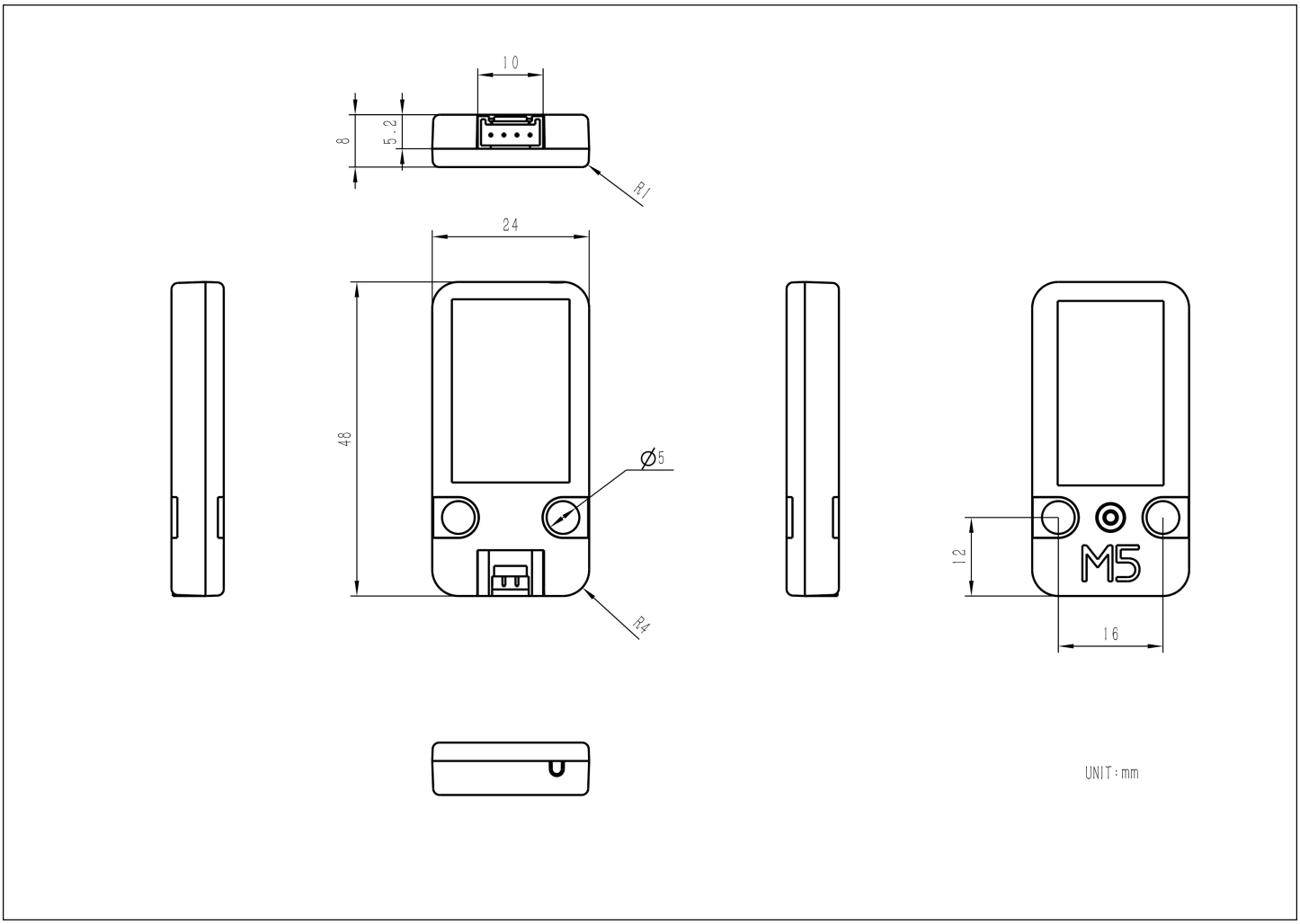
Unit UWB

HY2.0-4P	Black	Red	Yellow	White
PORT.C	GND	5V	UART_RX	UART_TX

Datasheets

- AT Command
- BU01-specification
- DW1000 Datasheet

Model Size



Structure

- UWB Unit Structure Files

Softwares

Arduino

- Unit UWB Arduino Tutorial

UiFlow1

- Unit UWB UiFlow1 Docs

UiFlow2

- Unit UWB UiFlow2 Docs

Protocol

Note: Each command must end with a carriage return and line feed **\r\n**

- Setting AT Commands

`AT+switchdis=` value

- Command description: Controls the ranging switch, only valid in tag mode
- Response:
  - When value equals 0, ranging is turned off, returns ok.
  - When value equals 1, ranging is turned on, returns ok.

`AT+interval=` value

- Command description: Sets the ranging interval
- Response: OK

> **Note:**
> The range is 5-50 meters, setting how many data points are collected before outputting the distance. The larger the value, the slower the refresh rate.

AT+version?

- Command description: Gets the manufacturer, module series, and version number
- Response: "AIT-BU01-DB V000 T2020-4-17 OK"

AT+RST

- Command description: Resets the module
- Response: OK

`AT+anchor_tag=` model,ID

- Command description: Sets the device mode and ID
- Parameters:
  - Model 1 selects anchor mode, 0 selects tag mode
  - ID sets the base station and tag ID
- Response: OK

```
// Base station mode
AT+anchor_tag=1,0
+anchor_tag=OK

AT+RST
+RST=OK
```

```
// Tag mode
AT+RST
+RST=OK

AT+anchor_tag=0
+anchor_tag=OK

AT+interval=5
+interval=OK

AT+switchdis=1
+switchdis=OK



// Returns ranging results
```

```
//
```

EasyLoader

Easyloader	Download Link	Notes
Unit UWB Example with M5Core	download	/

Video

UWB_VIDEO.mp4